

Yitong Bai

M.S. Student in Computer Engineering, Columbia University

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🌐 brianbai

Research Interests

- LLM agents, multi-agent systems, ML systems, evaluation and verification, reproducibility, trustworthy AI, and LLM post-training.

Education

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| M.S. Columbia University , Computer Engineering | New York, NY |
| • Expected Dec 2026 | Sept 2025 – Dec 2026 |
| • GPA: 3.83 / 4.00 | |
| B.S. (Hons) Lancaster University , Computer Science | Lancaster, UK |
| • First Class Honours; GPA: 3.73 / 4.00 | Sept 2023 – June 2025 |
| • BJTU-Lancaster Dual-Degree Program (2+2) | |
| B.S. Beijing Jiaotong University , Computer Science and Technology | Beijing, China |
| • GPA: 3.37 / 4.00 | Sept 2021 – June 2023 |
| • BJTU-Lancaster Dual-Degree Program (2+2) | |

Research Experience

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| Zhejiang Lab , Research Assistant | Hangzhou, China |
| Advisor: Jiaxi Yang, Research Scientist, Zhejiang Lab | June 2026 – present |
| <ul style="list-style-type: none">• Project: Feedback Learning for EHR SQL Repair: SFT vs. GRPO; manuscript in preparation.• Conducted a pre-registered study comparing SFT and GRPO on how diagnostic feedback shapes learning for EHR SQL repair, fine-tuning Qwen3.5-9B on MIMIC-IV EHR questions under four nested feedback levels, from a bare failure signal to repair hints.• Trained 12 QLoRA adapters (4 feedback conditions x 3 seeds, 1,200 samples) and evaluated semantic repair success across 13 model groups and 10,400 records, using paired comparisons with task-clustered confidence intervals, permutation tests, and Holm correction.• Found no stable incremental benefit from richer feedback under SFT: matched gains of +1.0 to +2.0 percentage points did not meet the pre-registered significance bar, a carefully bounded null result.• Built a target-free GRPO pipeline with a custom binary execution reward and observed a non-monotonic feedback effect, with the intermediate L2 level reaching +10.0 points (95% CI [+4.0, +16.5], exploratory Holm-adjusted $p = 0.0228$), pending multi-seed confirmation.• Ran all 12 adapters on a single 24 GB GPU under a strict no-leakage protocol, with reward verification isolated in a read-only database worker that never returns gold answers or verifier diagnostics to the model. | |

Columbia University, CRIS Lab, Research Assistant

Advisor: Prof. Venkat Venkatasubramanian

New York, NY

Sept 2025 – May 2026

- Project: DeepAudit: Agentic Evidence-Based Auditing of ML Reproducibility; manuscript in preparation.
- Designed and implemented DeepAudit, an agentic claim-level reproducibility auditing system that maps ML paper claims to executable repository tasks, metric contracts, runtime artifacts, and structured verdicts.
- Built a 23-step Python pipeline covering PDF-to-Markdown ingestion, table/figure-aware claim extraction, repository analysis, RAG-assisted code indexing, environment repair, autonomous execution, evidence alignment, scoring, and report generation.
- Formulated ML reproducibility auditing as claim-level evidence alignment rather than binary repository execution, distinguishing metric disagreement from missing data, incompatible environments, ambiguous entry points, unlogged metrics, and compute limits.
- Evaluated DeepAudit on 10 public ML paper-code pairs with 64 pre-specified audit targets; extracted 1,101 claims, identified 988 code-verifiable claims (89.7%), generated 49 metric contracts, and executed 52/64 experiment specifications (81.2%).
- Recovered executable or diagnostic evidence for all five initially broken repositories through bounded environment repair, autonomous execution, conservative verification, and structured failure analysis.

Columbia University / IBM Research Collaboration, Research Assistant

Mentored by Dhaval Patel, Senior Technical Staff Member, IBM Research

New York, NY

Feb 2026 – July 2026

- Project: Multi-Turn AssetOps: Supervisor-Specialist Agents for Industrial O&M.
- Proposed and designed a LangGraph-based Supervisor-Specialist multi-agent architecture for multi-turn industrial asset operations, routing user goals across data collection, time-series analysis, failure reasoning, and maintenance-planning specialists over MCP tools.
- Designed structured cross-turn artifact reuse to preserve sensor datasets, anomaly analyses, failure hypotheses, and maintenance recommendations as reusable evidence across follow-up turns.
- Iteratively optimized routing, artifact reuse, and MCP tool execution with domain-specific tool constraints, schema validation, duplicate-call suppression, and failure guards.
- Co-developed and evaluated the system on 16 multi-turn industrial O&M dialogs covering fault diagnosis, predictive maintenance, operational monitoring, maintenance planning, and full-pipeline remediation workflows.
- Reported improvements over a Plan-Execute baseline: +54.5% planning effectiveness, +37.8% task completion, tool-time share reduced from 47.3% to 26.3%, and follow-up turns approximately 4.2x faster after artifact reuse.
- Analyzed system-level trade-offs between specialist routing and parallel tool execution, showing that naive parallelism can reduce API calls while increasing token/context cost and tail latency.

SJTU Summer Campus, Project Leader

Facial Emotion Recognition with VGG16

Shanghai, China

July 2023 – Sept 2023

- Built a facial emotion recognition pipeline on the FER-2013 dataset using VGG16-based feature extraction and an MLP classifier.
- Applied data augmentation and feature fusion to improve robustness across emotion categories.
- Led a four-person team, coordinated task division and progress reviews, and completed both implementation and project reports.

Publications and Manuscripts

- Application and Analysis of the VGG16 Model in Facial Emotion Recognition** 2023
Yitong Bai
[10.5220/0012799800003885](https://arxiv.org/abs/10.5220/0012799800003885) (Proceedings of DAML 2023, pp. 422-426)
- Position: Agentic Science Needs Explicit Claim-Result-Computation Relationships** 2026
Yitong Bai
(Under review at AI for MetaScience NeurIPS Workshop)
- Beyond Static Leaderboards: Predictive Validity for the Evaluation of LLM Agents** 2026
Dhaval Patel, *Yitong Bai*, et al.
(Under review at ACL Rolling Review (ARR))
- The Label Is Not the Answer: Output-Schema Anchoring in Medical Multimodal MCQ Evaluation** 2026
Chengyi Zhang, *Yitong Bai*, Ziyang Wang
(Accepted at MICCAI 2026 Challenge MedReason)
- Towards Multi-Turn Dialog Systems for Industrial Asset Operations and Maintenance** 2026
Chengrui Li, Rujing Li, *Yitong Bai*, Rui Li
arxiv.org/abs/2605.24953 (arXiv preprint arXiv:2605.24953)

Industry Experience

- Neusoft Group**, Software Engineering Intern Guangzhou, China
July 2024 – Aug 2024
- Contributed to development and testing for a university portal system, including frontend refinement, database interface support, and cross-database data matching.
 - Worked with heterogeneous data sources to support synchronization and integration across subsystems.

Selected Projects

- Agentic Java Formal Verification** Sept 2025 – Dec 2025
Columbia University, Formal Verification of Hardware and Software Systems Project
- Developed a multi-agent prototype that connects LLM-based code understanding with Java / Spring Boot formal verification, translating source-code slices into candidate safety properties and JBM-ready harnesses.
 - Designed a three-stage pipeline with a code explorer agent, property formalizer agent, and harness generator agent, producing structured artifacts for parsed code slices, inferred properties, generated harnesses, and execution logs.
 - Tested the prototype on application-level properties including RBAC constraints, inventory-update consistency, and status-transition safety.
- ConvNeXt and ViT for Hierarchical Open-Set Image Classification** Sept 2025 – Dec 2025
Columbia University, Neural Networks and Deep Learning Project
- Built a two-stage hierarchical open-set recognition pipeline in PyTorch for super-class and sub-class prediction.
 - Implemented a unified classifier framework with interchangeable ImageNet-pretrained ConvNeXt and ViT backbones for controlled comparison.
 - Developed reproducible training utilities, including flexible data loading, automatic validation split, checkpointing, and metric logging.
 - Evaluated the approach on a 6,288-image dataset with 3 super-classes and 87 sub-classes.
 - On a held-out novel split, improved unseen sub-class recognition from 62.24% to 82.22% using ConvNeXt; compared Softmax-thresholding (91.60%) with OpenMax (59.60%) for novelty rejection.

TrueShot: P2P Blockchain Prototype for Photo Provenance Verification

Feb 2026 – May 2026

Columbia University, Computer Networks Course Project

- Built a peer-to-peer photo provenance prototype with tracker-based peer discovery, gossip-based transaction and block propagation, a proof-of-work blockchain, Merkle-root validation, ECDSA-signed device records, and a Flask verification UI.
- Implemented security demos including forged longest-chain replacement, tracker poisoning, gossip sender URL spoofing, mempool flooding, and image tampering detection; the project report documents 69 passing tests across blockchain, network, application, and web layers.

Computer Vision and ML Lead, Pet Companion Robot

Sept 2025 – Dec 2025

Columbia University, AIoT Project

- Led the computer vision and machine learning components of an AIoT pet robot integrating on-device perception, motion control, cloud analysis, and dashboarding.
- Built real-time dog detection and motion control on Raspberry Pi 5 using YOLO, FastAPI, sensor fusion, and target tracking.
- Developed cloud pipelines for automated image capture, behavior summarization, and a VGG16-based pet emotion classifier exposed through reusable APIs.

Honors and Awards

- Honorable Mention, Mathematical Contest in Modeling, COMAP (Mar 2024)
- Beijing Jiaotong University Scholarship for Academic Progress, Top 5% (Jun 2022)

Skills

Programming Languages: Python, C, Java, R, SQL

LLM Agents: LangGraph, LangChain, RAG, MCP / FastMCP, LiteLLM, Pydantic, OpenAI API

LLM / ML: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, LLaMA-Factory, TRL, QLoRA, SFT, GRPO, vLLM, Weights & Biases

Evaluation & Reproducibility: pre-registered experiment protocols, paired comparisons with task-clustered confidence intervals, permutation tests, multi-seed replication

Systems / Engineering: Linux, Docker, Git, Conda / mamba / venv, pytest, FastAPI, Flask